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TradeLENS

CNN-Based Trading Strategy with Integrated Technical and Macroeconomic Representations

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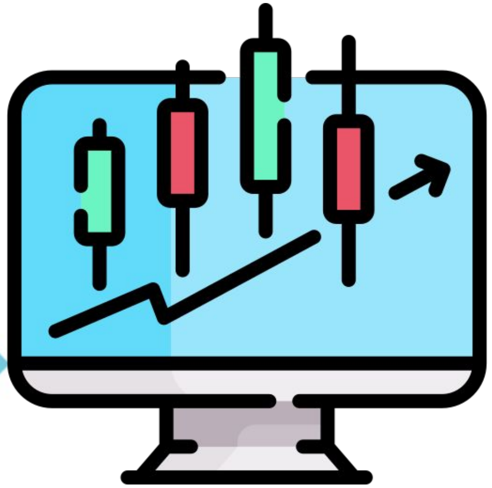
INTRODUCTION



Financial markets are driven by price, indicators, and macroeconomics.

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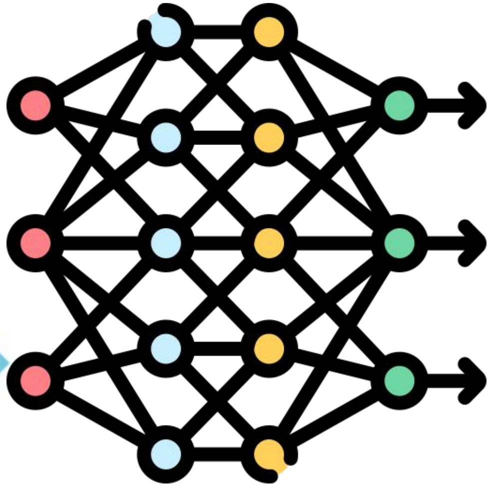

CNNs extract complex spatial features from images to enable accurate and efficient image classification.

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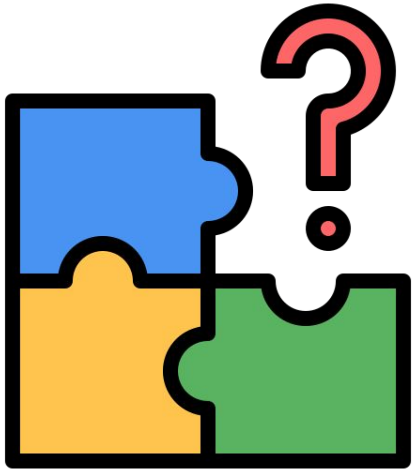

Existing CNN models are usually trained on technical indicators only, and they rarely encode macroeconomic context in the same input representation.

STATEMENT OF THE PROBLEM

1. Do macroeconomic indicators improve the performance of CNN-based trading models when combined with technical indicator images?

2. How do macroeconomic indicators contribute to the accuracy and effectiveness of CNN-based trading systems?

METHODOLOGY

Dataset

Historical Stock Price Data

Macroeconomic Indicators

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METHODOLOGY

Dataset

Historical Stock Price Data

Open Price
High Price
Low Price
Close Price

From 2012 to 2025

METHODOLOGY

Dataset

1. Reverse repurchase rate (RRP)
2. Purchasing power of peso
3. Oil Price
4. Gold Price
5. Inflation rate

Macroeconomic Indicators

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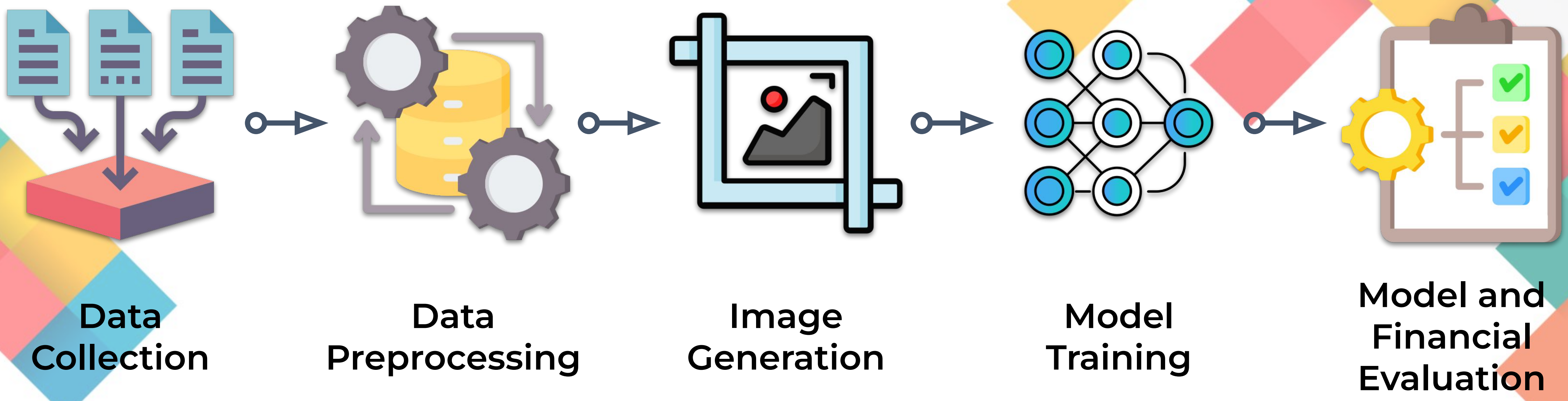
METHODOLOGY

TradeLENS Process Flow



METHODOLOGY

Data Preprocessing

1. Relative Strength Index (RSI)
2. Williams %R
3. Simple Moving Average (SMA)
4. Exponential Moving Average (EMA)
5. Weighted Moving Average (WMA)
6. Hull Moving Average (HMA)
7. Triple Exponential Moving Average (TEMA)
8. Commodity Channel Index (CCI)
9. Chande Momentum Oscillator (CMO)
10. Moving Average Convergence Divergence (MACD)
11. Percentage Price Oscillator (PPO)
12. Rate of Change (ROC)
13. Chaikin Money Flow Indicator (CMFI)
14. Directional Movement Indicator (DMI)
15. Parabolic Stop and Reverse Indicator (PSI)

METHODOLOGY

Data Preprocessing

Algorithm 1 Turning point label generation (Buy, Hold, Sell)

```
procedure LABELGEN_TURNINGPOINT
    N = length(close_prices)                ▷ close_prices = Daily close price series
    R ← 5                                   ▷ 11-day window, 5 days before and after
    for t = 0 to N − 1 do
        start ← max(0, t − R)
        end ← min(N − 1, t + R)
        window ← close_prices[start..end]
        center_price ← close_prices[t]
        max_price ← max(window)
        min_price ← min(window)
        max_count ← number of positions in window equal to max_price
        min_count ← number of positions in window equal to min_price
        if center_price = max_price and max_count = 1 then
            label_text[t] ← "Sell"
            label_code[t] ← −1
        else if center_price = min_price and min_count = 1 then
            label_text[t] ← "Buy"
            label_code[t] ← 1
        else
            label_text[t] ← "Hold"
            label_code[t] ← 0
        end if
    end for
    save table with columns: dates, close_prices, label_text, label_code
    return label_code
end procedure
```

METHODOLOGY

Data Preprocessing

20 × 20 matrix

Label: BUY, SELL, HOLD

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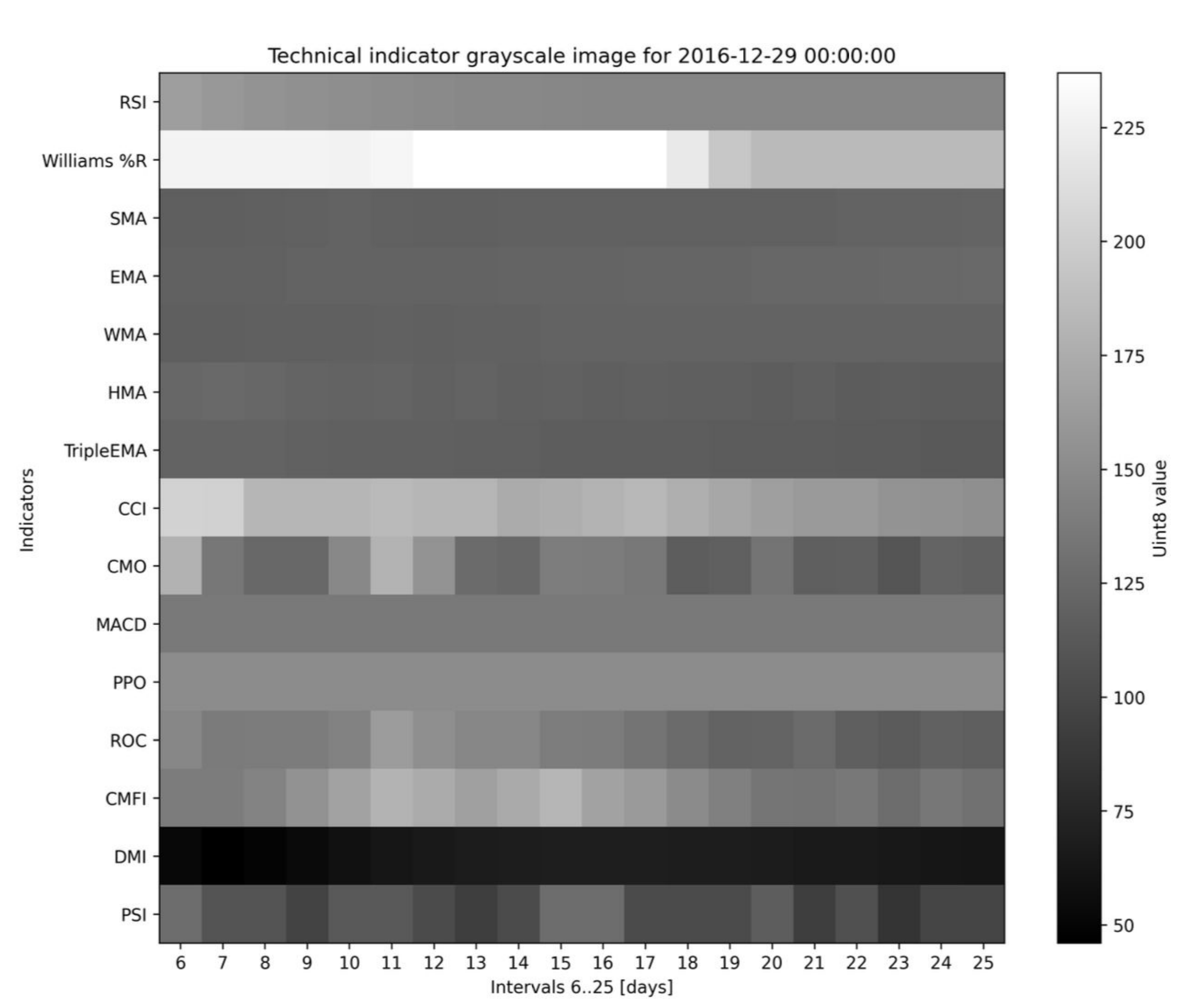
METHODOLOGY

Image Creation



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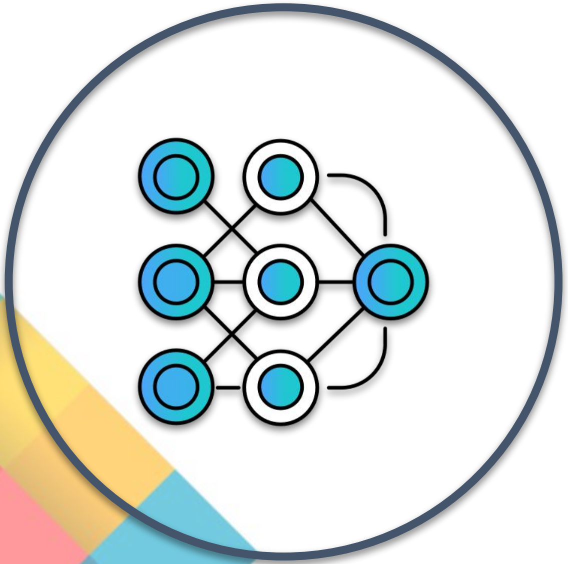
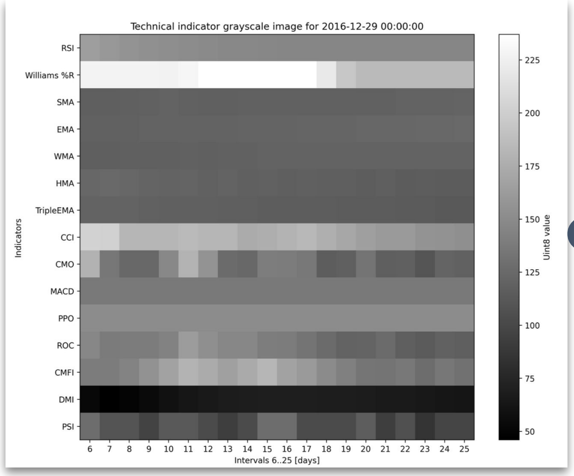
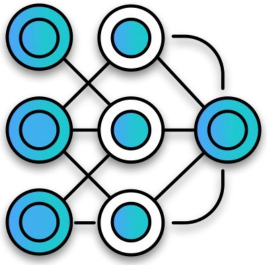
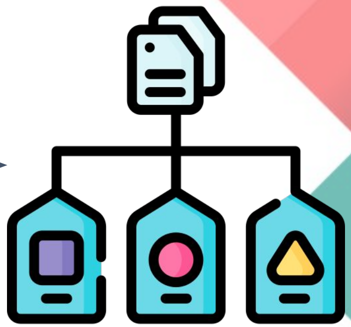


Model Training







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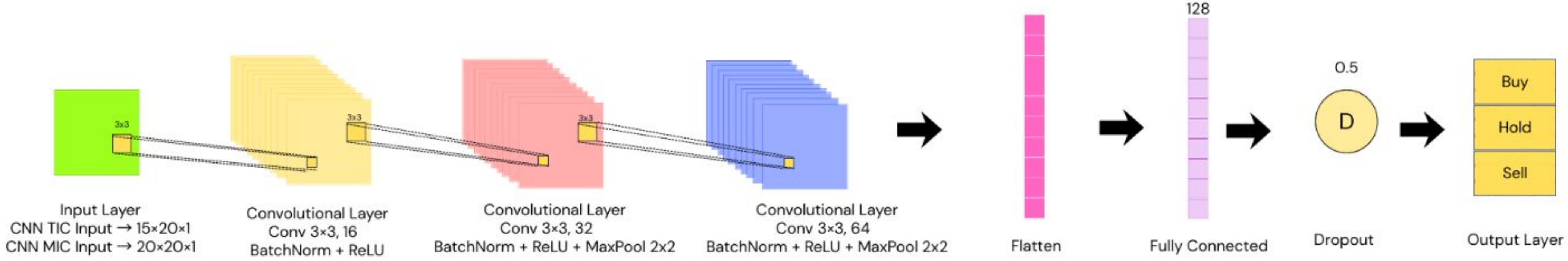


METHODOLOGY

Model and Financial Evaluation

Model Evaluation

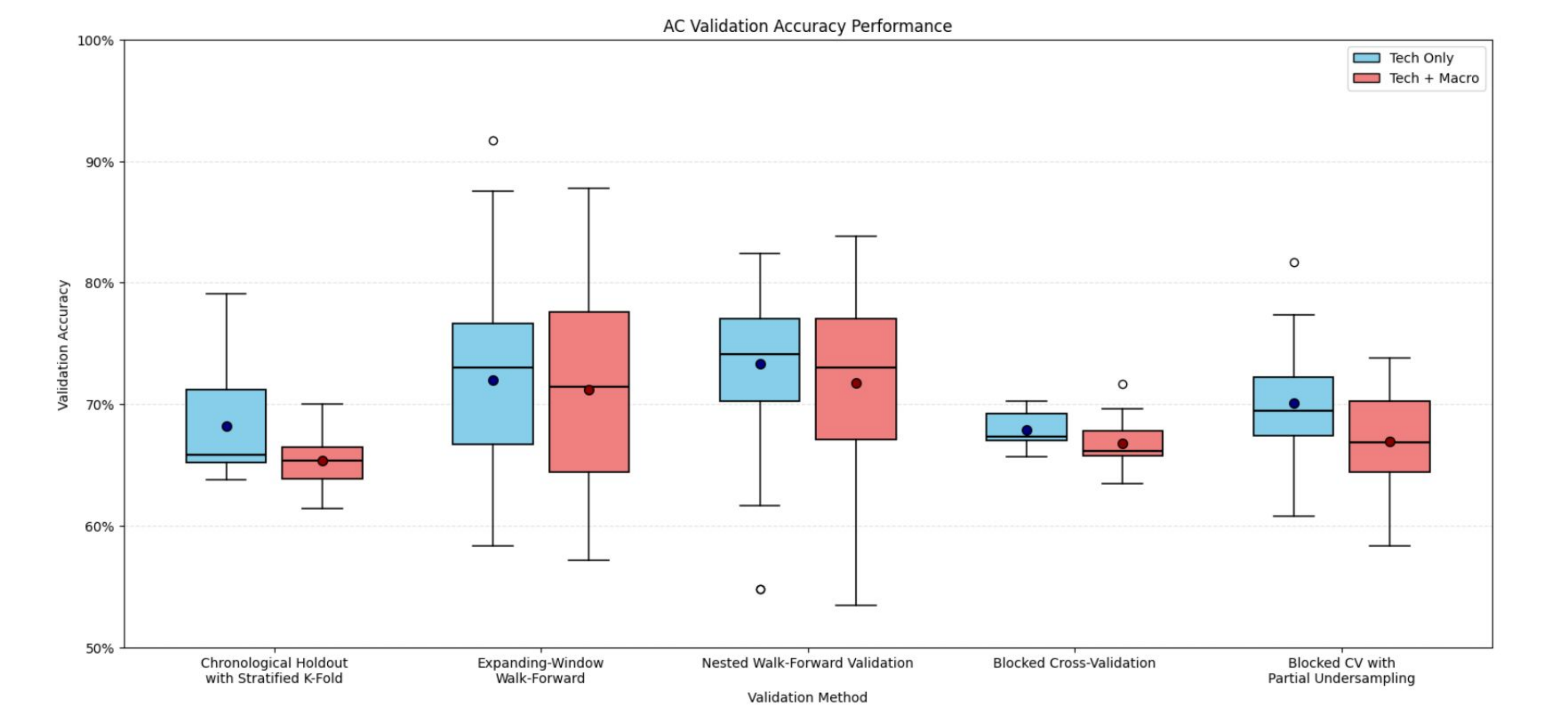
1. Accuracy
2. Precision
3. Recall
4. F1-score

Financial Evaluation

- Trading Simulation
- Php 10,000.00 initial equity

RESULTS AND DISCUSSIONS

Integrating technical and macroeconomic indicators did not significantly improve the classification performance of the trading model.



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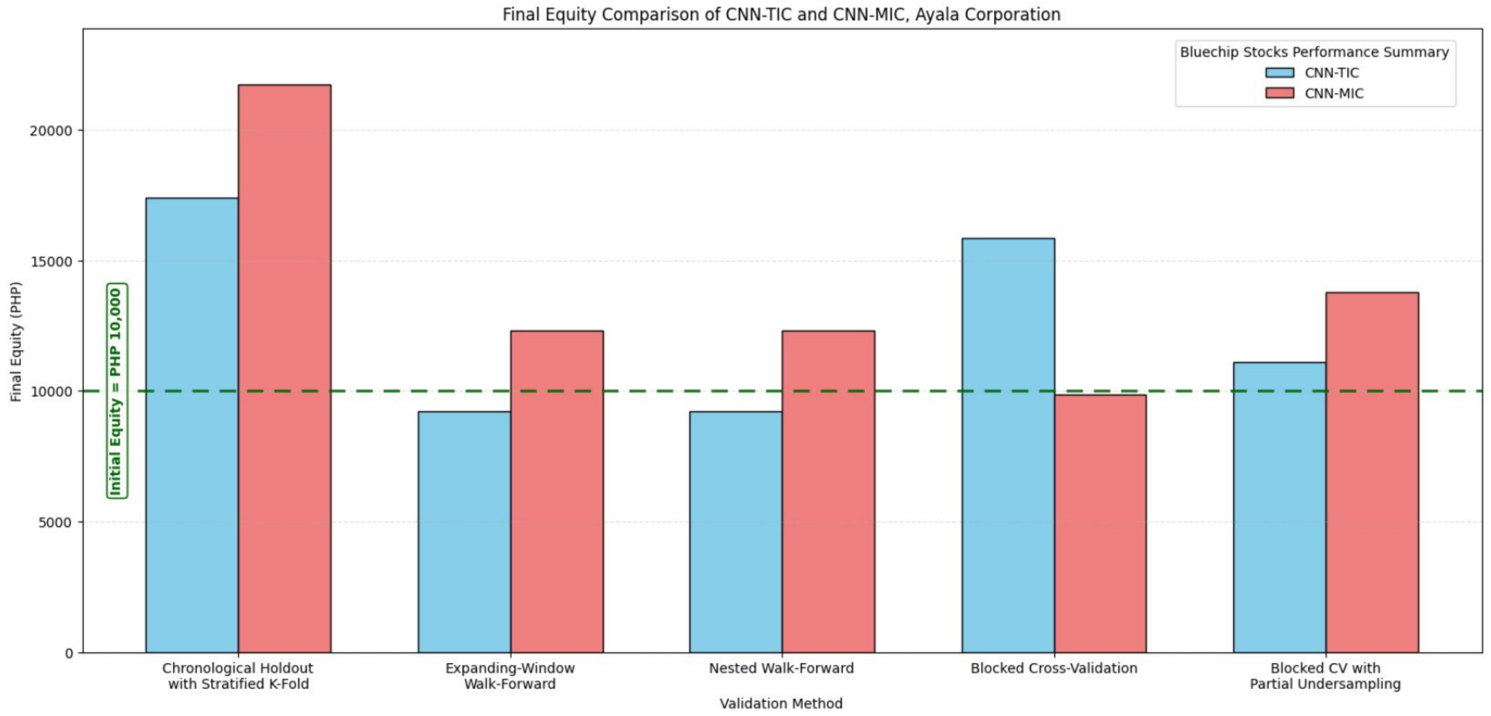
RESULTS AND DISCUSSIONS

Trading simulations demonstrated that integrating technical and macroeconomic indicators significantly enhanced the financial performance of the model.



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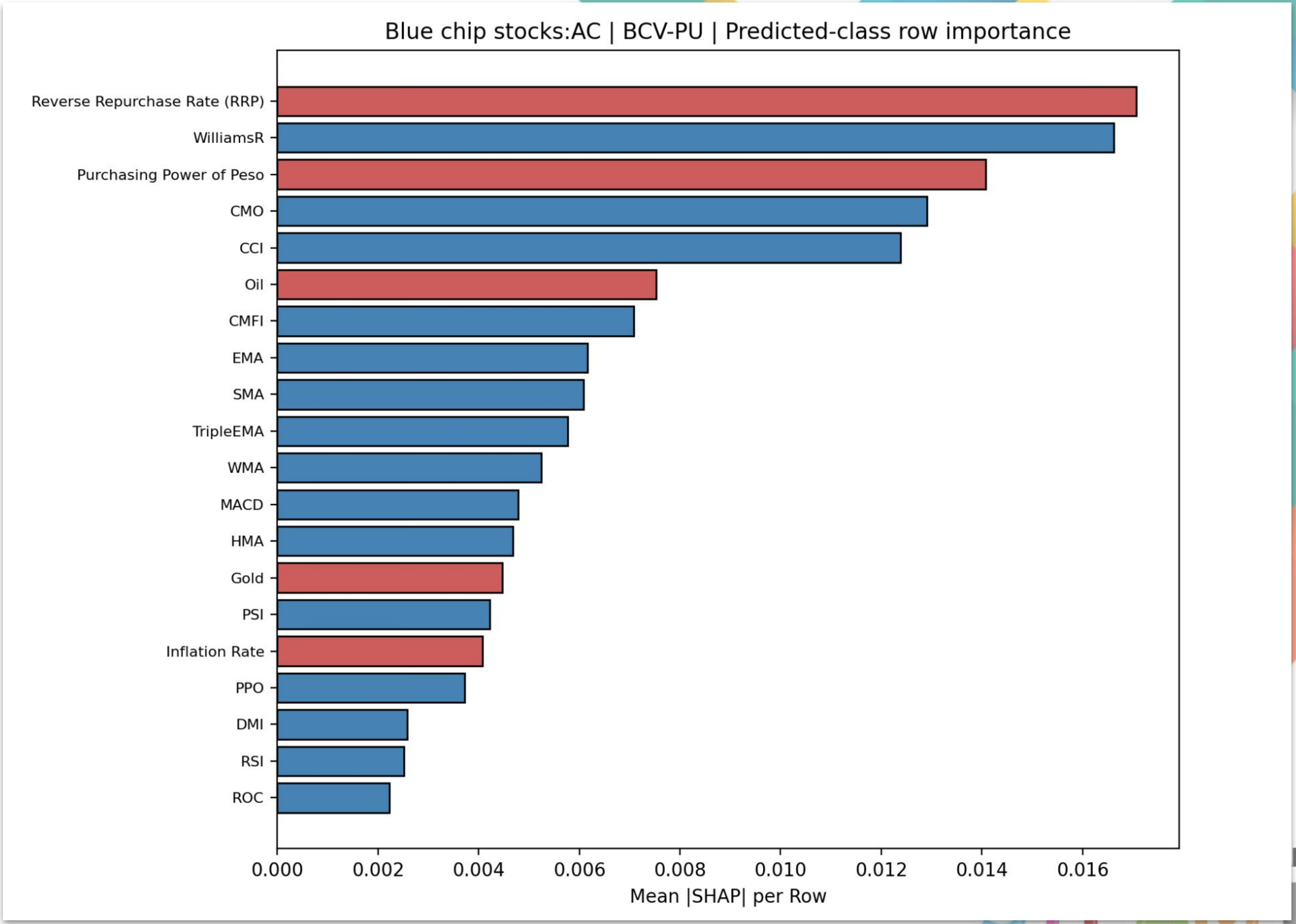
RESULTS AND DISCUSSIONS

Feature analysis revealed that certain macroeconomic indicators possess high feature importance and effectively complement technical indicators, thereby improving the overall performance of the trading model.



CONCLUSION

TradeLENS integrates technical and macroeconomic indicators to enhance trading model analysis.

The inclusion of macroeconomic indicators did not significantly improve classification accuracy.

Trading simulations demonstrated improved financial returns using the integrated feature set.

Feature importance analysis showed that macroeconomic indicators effectively complemented technical indicators in supporting trading model performance.

TradeLENS: CNN-Based Trading Strategy with Integrated Technical and Macroeconomic Representations

THANK YOU

Questions?

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